

YUANSHUAI LI

Shenzhen, China — +86 188 3970 5185 — 122090293@link.cuhk.edu.cn

Research Interests

Multimodal Large Language Models; Reinforcement Learning; Deep Learning;

Education

The Chinese University of Hong Kong, Shenzhen

Shenzhen, China

*B.Eng. in Computer Science and Engineering, **First Class Honours***

Sep 2022 – Jul 2026

- **GPA:** 3.699/4.000.
- **Honours:** Dean's List for three consecutive academic years (2022–2025); Entrance Scholarship.
- **Selected coursework:** Reinforcement Learning (A), Machine Learning (A-), Advanced Machine Learning (A), Design and Analysis of Algorithms (A), Database System (A), Probability (A).

Research Experience

In-Context Reinforcement Learning

Shenzhen, China

Research Assistant, supervised research project

Jun 2025 – Aug 2025

- Authored an 11-page survey on supervised-pretraining approaches to in-context reinforcement learning, synthesizing Algorithm Distillation, DPT, IDT, and ICEE across training objectives, exploration, adaptation, and scalability.
- Reproduced the Decision-Pretrained Transformer (**DPT**) in Google Colab and ran small-scale experiments, gaining hands-on experience with trajectory construction, Transformer-based decision policies, and empirical evaluation.
- Identified open problems in long-context computation, data efficiency, exploration, and theoretical guarantees for in-context reinforcement learning.

BIRD2025 / SWE-SQL LLM–Database Project

Shenzhen, China

Research Assistant

Oct 2024 – Feb 2025

- Helped construct a benchmark for evaluating how large language models solve real-world database problems across MySQL, PostgreSQL, and Oracle.
- Curated, solved, and validated **100+** challenging Stack Overflow cases spanning diverse error types and complex SQL scenarios; designed executable tests to assess the correctness of model-generated solutions.
- **Co-author:** Jinyang Li, Xiaolong Li, Ge Qu, . . . , **Yuanshuai Li**, et al. “SWE-SQL: Illuminating LLM Pathways to Solve User SQL Issues in Real-World Applications.” *NeurIPS 2025 Main Conference*.

Data Space Testbed Development

Shenzhen, China

Core Developer

Jun 2024 – Sep 2024

- Studied International Data Spaces Association specifications and analyzed the interactions among DAPS, certificate authority, and connector components.
- Built a Python/Docker testing environment and designed authentication and authorization tests for DAPS, contributing to an end-to-end testbed for core data-space components.

Technical Skills

Programming: Python, Java, C++ **ML:** PyTorch, TensorFlow, Transformer and reinforcement-learning fundamentals

Tools and Systems: Linux, Docker, Git, Google Colab **Data:** MySQL, PostgreSQL, SQL benchmarking

Languages: Mandarin (native), English (CET-4 and CET-6; academic reading and writing)